

UGED1111 Logic

CATEGORICAL SYLLOGISM II

CHAN, Hei Man

Agenda

- **Test the validity with Venn diagram**
- **Translating Ordinary Language Statements into Categorical Form**

Categorical Syllogism

A CATEGORICAL SYLLOGISM is a
deductive argument with:

- a. 3 categorical propositions: two premises and one conclusion.**
- b. 3 terms, each occurring twice .**
- c. Each term occurring in two statements.**

Categorical Syllogism

- The MAJOR TERM is the predicate term (P) of the conclusion.
- The MINOR TERM is the subject term (S) of the conclusion.
- The MIDDLE TERM is the term (M) occurring in both premises but not in the conclusion.
- *The MAJOR PREMISE is the premise containing the major term.*
- *The MINOR PREMISE is the premise containing the minor term.*

STANDARD FORM

STANDARD FORM:

1. Major Premise
 2. Minor Premise
 3. Conclusion
-

Example

1. All M are P.
 2. Some M are S.
 3. All S are P.
-

Check the Validity I

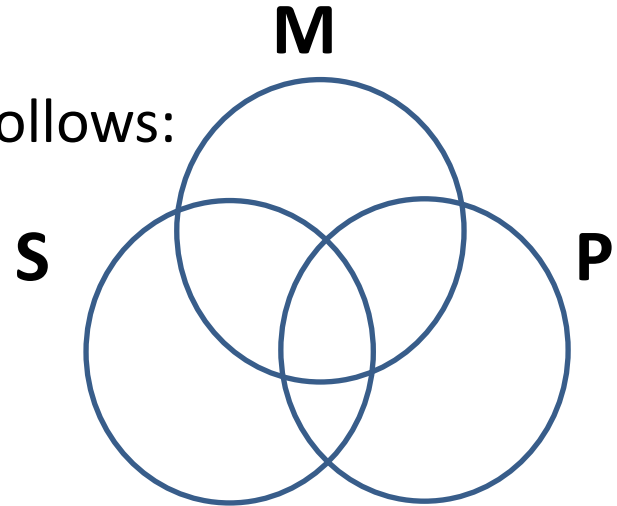
We can check the validity of Categorical Syllogisms with Venn diagrams

1. Draw three circles and label them as follows:

P is the Major term.

M is the Middle term,

S is the Minor term,

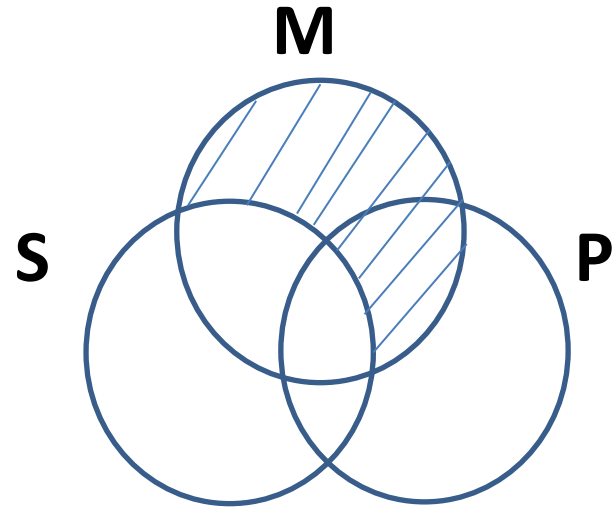


Check the Validity II

2. Marks (shading or placing an X) are entered only for the premises. No marks are made for the conclusion.

3. **Universal premises** should be marked on the diagram **first** and then particular premises. No priority of premises marking if both premises are universal or both are particular.

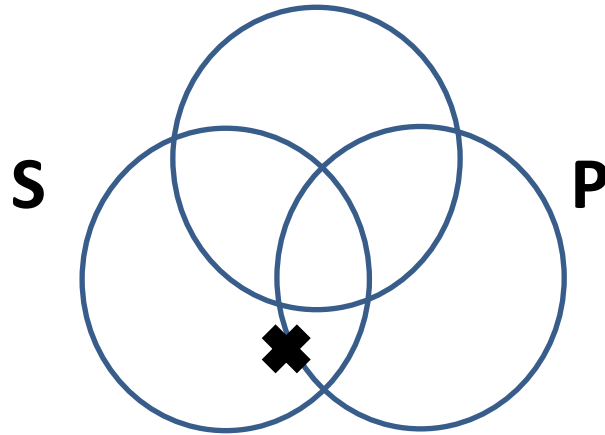
4. Shade the area where no objects can exist for the universal premises



e.g. All M are S.

Check the Validity III

5. Put an X in the area where at least one object exists. If it is possible to put an X in two areas, then put it on the borderline between the two regions **M**



e.g. Some S are not M.

6. Check if the conclusion is represented on the diagrams. If Yes, then the syllogism is **VALID**. If No, it is **INVALID**.

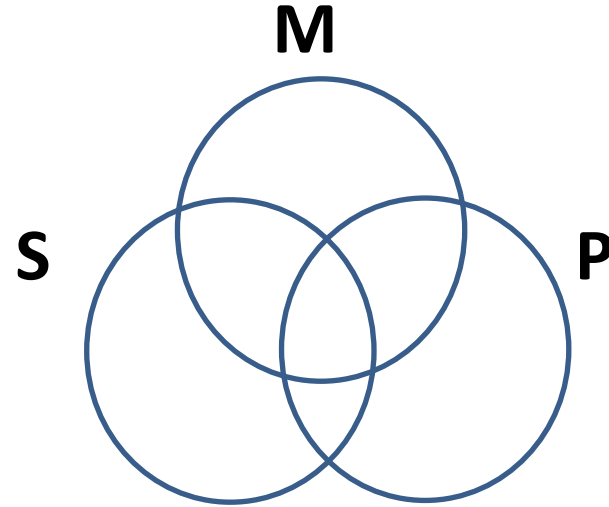
Recap

STANDARD FORM:

1. Major Premise

2. Minor Premise

3. Conclusion



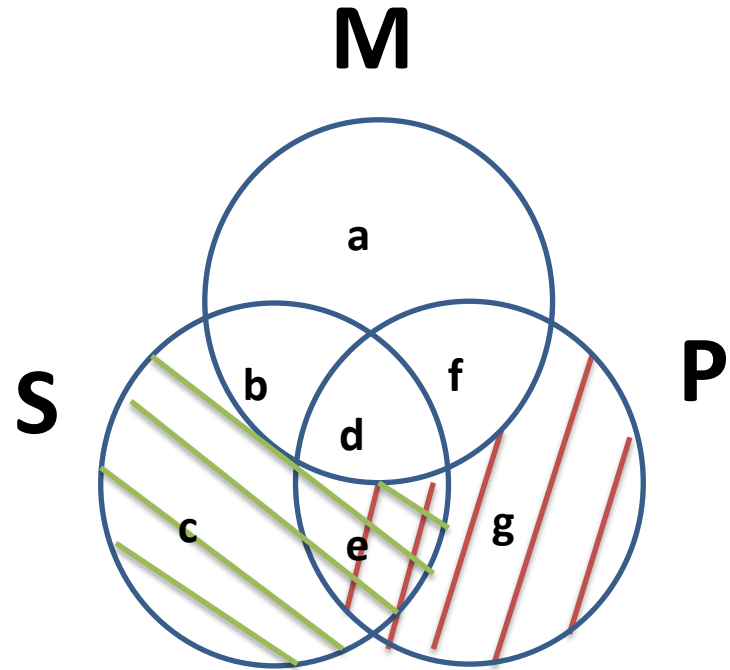
*Mark Universal premises on diagram first, then particular premises.

** No marks are made for the conclusion.

Example 1

1. All P are M.
 2. All S are M.
 3. No S are P.
-

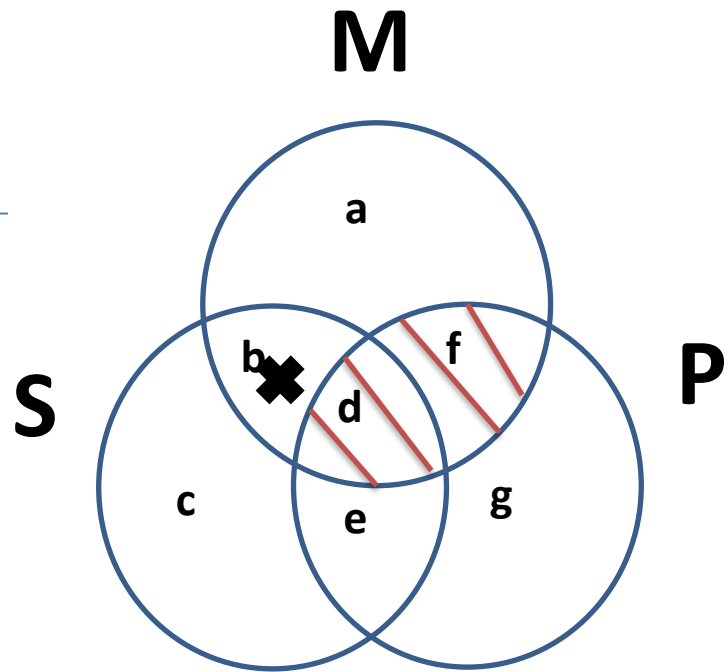
Invalid



Example 2

1. No P are M.
 2. Some S are M.
-
3. Some S are not P.

Valid



Existence

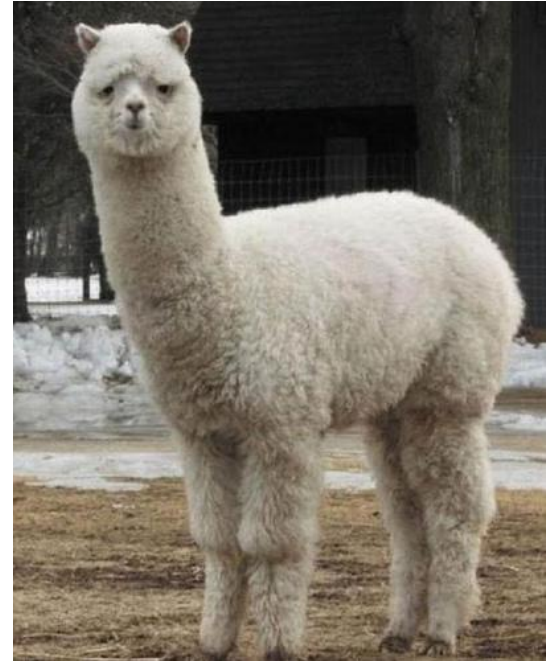
Some : there exists/is at least one

What does it mean by “exist”?

Sheep? IRONMAN?

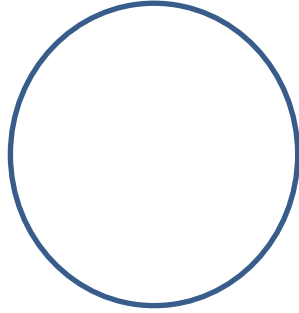
草泥馬? Unicorn?

Round-square?

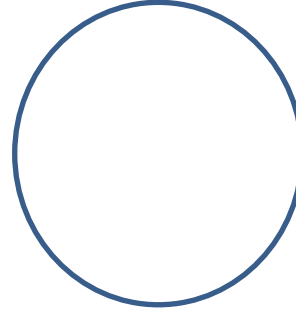


Existence

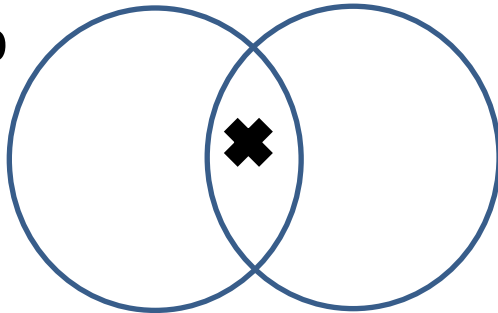
Sheep



Black
objects



Sheep



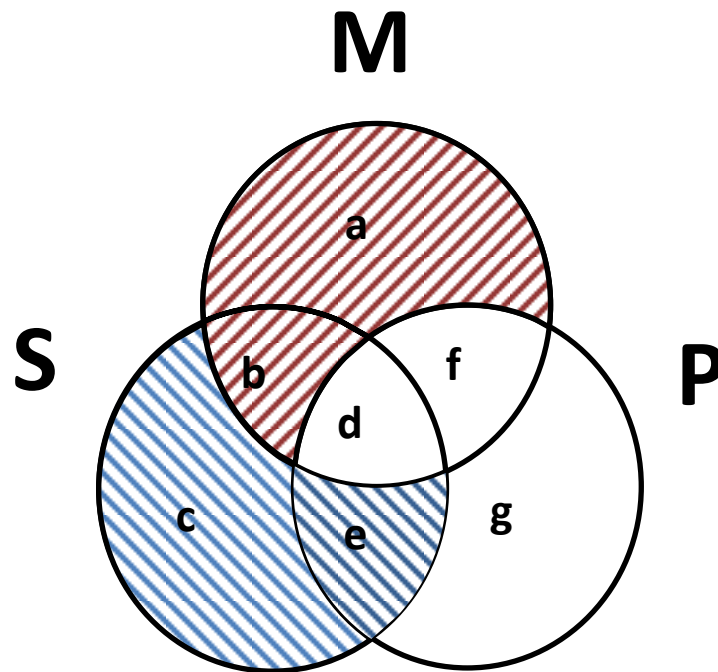
Black
objects

Some Sheep are black objects.

Discussion 1

Determine whether the following argument is valid or not

1. All M are P.
 2. All S are M.
-
3. All S are P.



Valid

Discussion 2

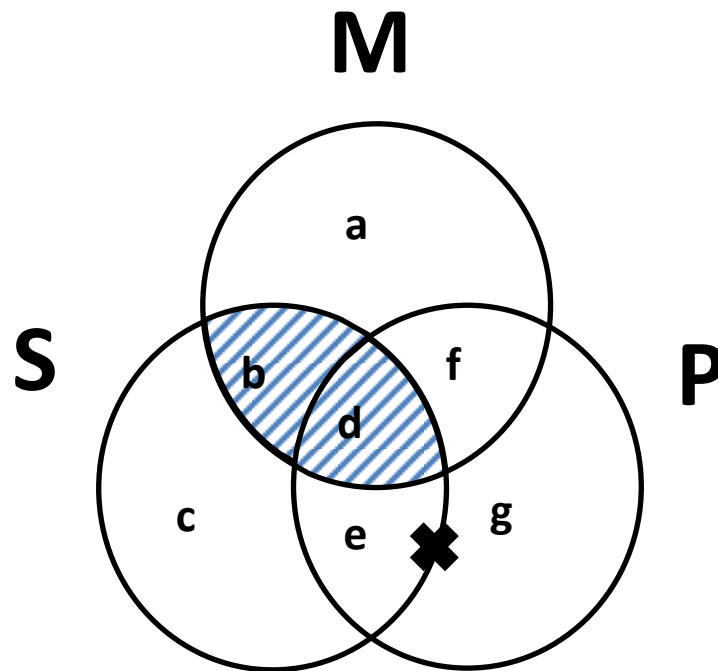
Determine whether the following argument is valid or not

1. Some P are not M.

2. No S are M.

3. Some S are not P.

Invalid



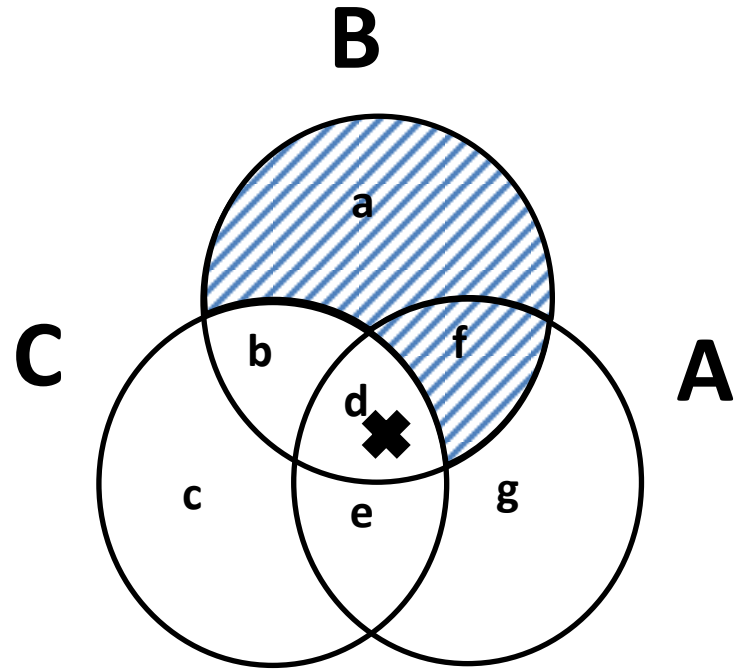
Discussion 3

Determine whether the following argument is valid or not

No non-C are B. Not all A are not B
Therefore, some C are A.

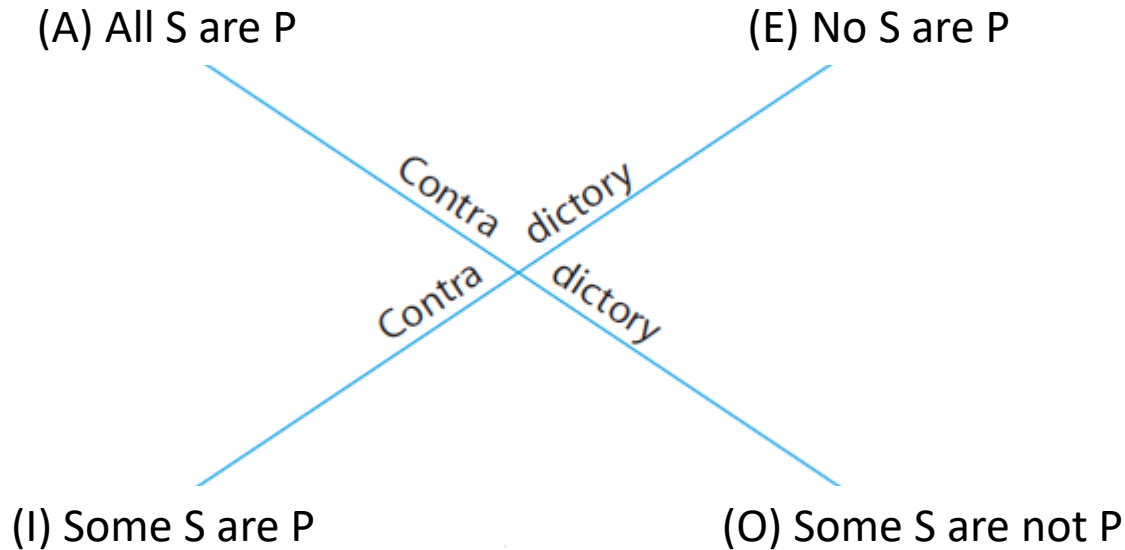
1. Some A are B.
2. All B are C.
3. Some C are A.

Valid



Recap

Modern Square



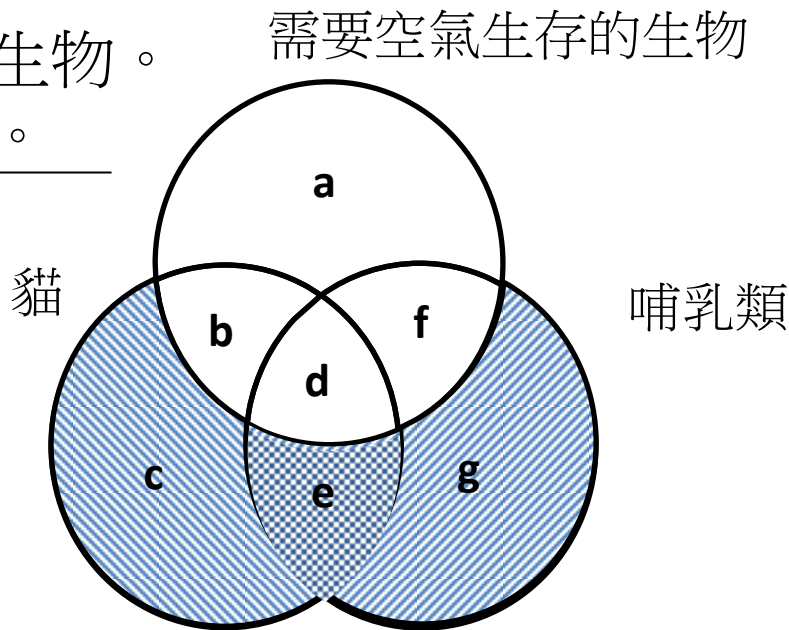
**** “All” does not imply “some”**

Example

Determine whether the following argument is valid or not

1. 所有哺乳類都是需要空氣生存的生物。
 2. 所有貓都是需要空氣生存的生物。
-
3. 所有貓都是哺乳類。

Invalid



Translation

General rule

• Understand the meaning of the given statement, and then re-express it in a new statement that has a **quantifier, subject term, copula, and predicate term.**

Standard quantifier: All ; No; Some

Copula: are; are not

Subject term and predicate term must be a noun.

Terms Without Nouns

If a term consists of only an adjective, a noun should be introduced to make the term genuinely denotative.

Example:

“All swans are **white**.”

- “All swans are **white animal**.”

Nonstandard Verbs

The only copulas that are allowed in standard-form categorical propositions are "are" and "are not."

Statements in ordinary usage often incorporate other forms of the verb "to be."

Example:

"Some staffs **will get year-end bonus.**"

- "Some staffs **are persons who will get year-end bonus.**"

Singular Proposition

Categorical logic concerns categories.

How about the proposition like:

"Socrates is mortal"

- "All persons identical to Socrates are persons who are mortal."

Singular propositions are typically translated into universal by means of a parameter.

Why do we use “all” instead of “Some” as in such cases?

Adverbs and Pronouns

- When a statement contains a **spatial** adverb such as "where," "wherever," "anywhere," "everywhere," or "nowhere," or a **temporal** adverb such as "when," "whenever," "anytime," "always," or "never,"

It may be translated in terms of **"places"** or **"times"** respectively.

Adverbs and Pronouns

Statements containing pronouns such as "who", "whoever", "anyone", "what" "whatever" or "anything" may be translated in terms of "persons" or "things" respectively.

Examples:

1. "Whoever works hard will succeed."
 - "All persons who work hard are persons who will succeed."
2. "Nowhere are good without logic."
 - "No good places are places without logic."

Unexpressed Quantifiers

In introducing the quantifiers one must be guided by the most probable meaning of the statement.

Example:

1. “There are monkeys in CUHK.”
 - “Some monkeys are animals in CUHK.”
2. “A few lecturers are nice.”
 - “Some lecturers are nice persons.”

Conditional Statements

When the antecedent and consequent of a conditional statement talk about the same thing, the statement can usually be translated into categorical form. Such statements are always translated as universals.

“If a person drinks red bull, then he will have wings.”

- “All persons who drink red bull are persons who have wings.”

‘only if ’

“An animal is a fish only if it lives in water.”

- “All fishes are animals that live in water.”

“Unless”

“Unless” means if not

“Tomatoes are edible unless they are spoiled.”

- “All unspoiled tomatoes are edible tomatoes.”

“Students will fail this course unless they attend the exam.”

- “All students who do not attend the exam are students who fail this course.”

Statements containing “unless” are translated as categorical propositions having negated subject terms.

*Usually we avoid negative terms

“Unless the sun is shining, a rainbow cannot occur.”

- “All times that a rainbow occurs are times that the sun is shining.”

Exclusive Propositions

Many propositions that involve the words "only," "none but," "none except," and "no ...except" are exclusive propositions.

Example

“Only nuclear-powered vehicles are suitable for deep-space exploration.”

- “All vehicles which are suitable for deep-space exploration are nuclear-powered vehicles .”

Exclusive Propositions

- “None but hard-working students will attend 8:30 lessons.”
“All students who attend 8:30 lessons are hard-working students “

language following "only," "none but " "none except," and "no ... except" goes in the predicate term of the categorical proposition.

“The Only”

Propositions that involve the words “the only,” are different from “only”

Example

“The only mammals which lay eggs are Platypus”.

- “All egg-laying mammals are Platypus.”

Language following “the only” goes in the subject term of the categorical proposition.

Pragmatic Meaning

Semantics Vs Pragmatic

“你的桌上是不是有胡椒粉?”

Context- dependent

***For exam, we translate the statement based on the literal meaning only. For daily usage , we should also understand the meaning under that particular context.**

Discussion 4



1. **“Not everyone who celebrate Valentine is in love.”**
“Some persons who celebrate Valentine are not persons in love.”
2. **“No birds excepts penguin can swim.”**
“All birds that can swim are penguins.”

Discussion 4



3. “It is not the case that nobody loves geeks.”

“Some persons are persons who loves geeks.”

4. “An argument is not sound unless it is valid”

“All sound arguments are valid arguments”

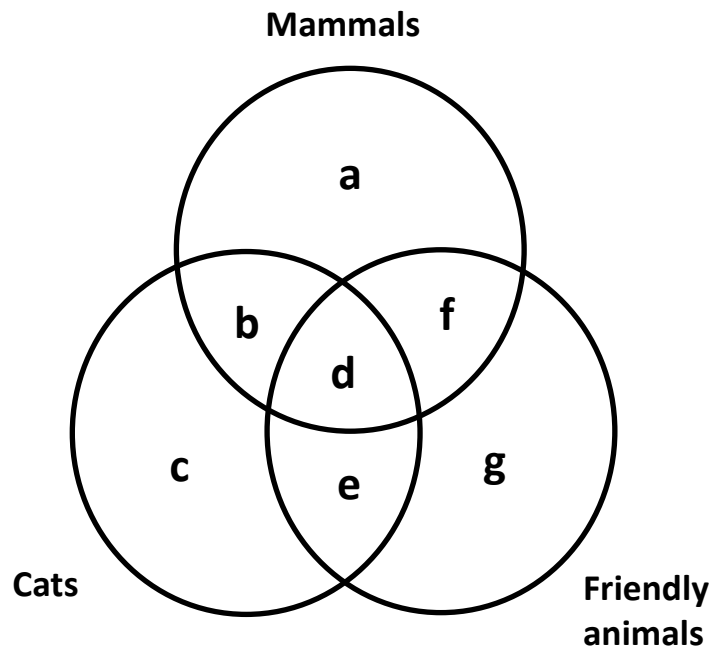
5. “No Hong Kong people will get married except who have owned a flat.”

“All Hong Kong people who get married are Hong Kong people who have owned a flat. “

Discussion 5

If it is not a mammal, it is not a cat. A few mammals are friendly. Therefore, Some cats are friendly animals.

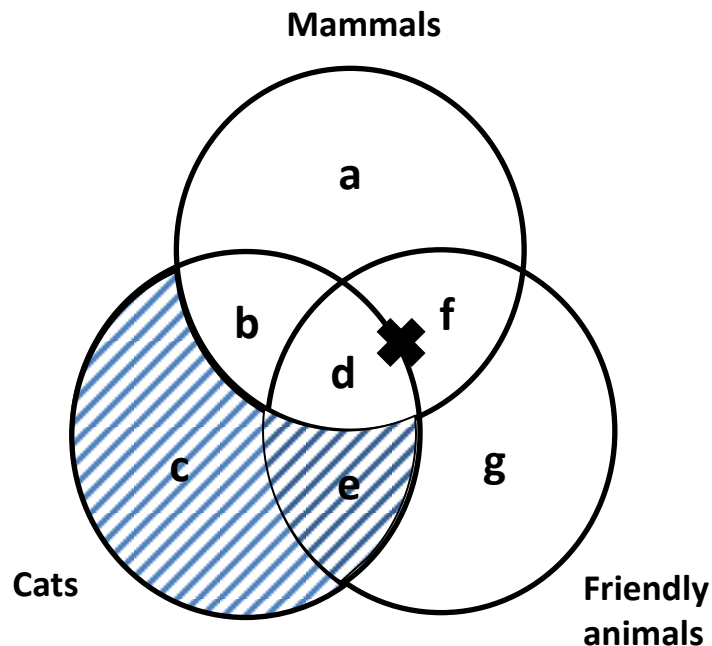
- 1. Some mammals are friendly animals.**
 - 2. All Cats are mammals.**
-
- 3. Some cats are friendly animals.**



Discussion 5

1. Some mammals are friendly animals.
 2. All Cats are mammals.
-
3. Some cats are friendly animals.

Invalid



Discussion 6

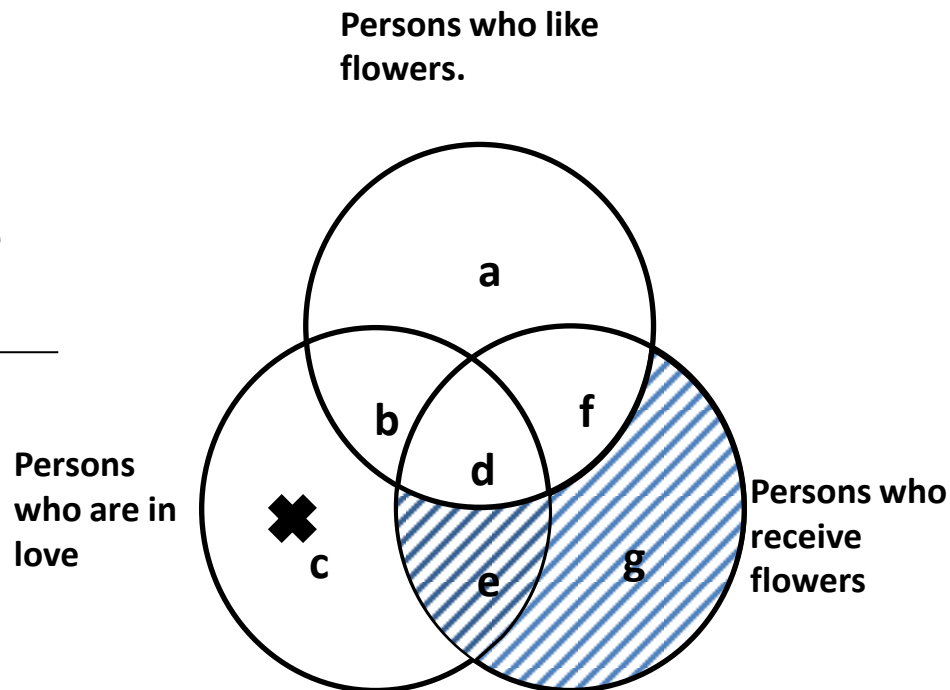
It is false that everyone in love like flowers. Only who like flowers will receive flower. Therefore, it is not the case that everyone in love will receive flowers.

- 1. All person who receive flowers are persons who like flowers.**
- 2. Some persons in love are not persons who like flowers.**
- 3. Some persons in love are not persons who receive flowers**

Discussion 6

1. All persons who receive flowers are persons who like flowers.
 2. Some persons who are in love are not persons who like flowers.
-
3. Some persons who are in love are not persons who receive flowers

Valid



Translation in Categorical Logic

Singular propositions are typically translated into universal by means of a parameter.

Why do we use “all” instead of “Some” as in such cases?

Consider , “Socrates is handsome.”

If we try to translate it into a “some “ statement, then it will be “Some persons are persons who are handsome”

Does this translation express the meaning of “Socrates is handsome.” better than "All persons identical to Socrates are persons who are handsome." ?

Translation in Categorical Logic



Even though we try our best to translate in a way to keep the original meaning of statements in ordinary language, sometimes it is hard for us to retain the meaning of the original statement.

Case

Consider the validity of the following arguments:

- | | |
|---|----------------------------|
| 1. 如果上帝無法造出自己舉不起來的石頭，
上帝不是全能的。 | 1. $\sim P \supset \sim Q$ |
| 2. 如果上帝可以造出自己舉不起來的石頭，
上帝不是全能的。 | 2. $P \supset \sim Q$ |
| 3. 上帝可以造出自己舉不起來的石頭或上
帝無法造出自己舉不起來的石頭。 | 3. $P \vee \sim P$ |
| <hr/> | |
| 4. 上帝不是全能的。 | // $\sim Q$ |

Problem of Categorical Logic

Some good arguments in natural language can not be proved in categorical logic.

- 1. All cats are mammals.**
- 2. All cats' tails are mammals' tails.**

We will consider it as a good argument. However, we cannot prove it in the system of Categorical Logic.

Problem of Categorical Logic

1. All cats are mammals.
 2. All cats' tails are mammals' tails.
-

Categorical logic can only handle arguments with 3 terms.
However, the argument involves 4 terms.

The expressive power of categorical logic is low.

Reference

- Copi, I., Cohen, C., & McMahon, K. (2011). *Introduction to Logic (14th ed. / Irving M. Copi, Carl Cohen, Kenneth McMahon ed.)*. Upper Saddle River, NJ: Pearson Education.
- Hurley, P. (2015). *A Concise Introduction to Logic (12th ed.)*. Australia ; Stamford, Ct.: Cengage Learning.
- Lau, J. (2011). *An Introduction to Critical Thinking and Creativity : Think More, Think Better*. Hoboken, N.J.: Wiley.